Homework 7

EE 999 — Introduction to Control Systems

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Due Wednesday, November 4, 2026

Problem 1 — Review questions 4.1 and 4.2

4.1

Feedback can:

- reduce the steady state error inherent to a system
- reduce the impact of disturbances to a system
- provide stability to unstable systems

4.2

Feedback typically requires:

- more logical complexity
- physical sensors (cost association)

Problem 2 — Text problem 4.6

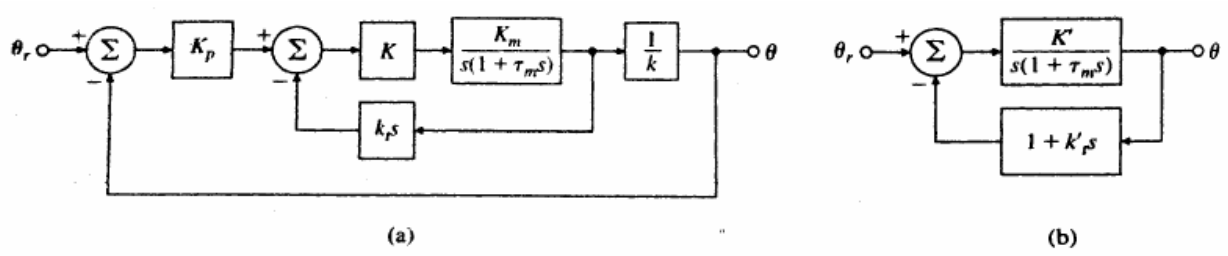


Figure 4.26
Control system for Problem 4.6

Figure 1: Text figure 4.26 (8th Ed.).

Problem 2.A

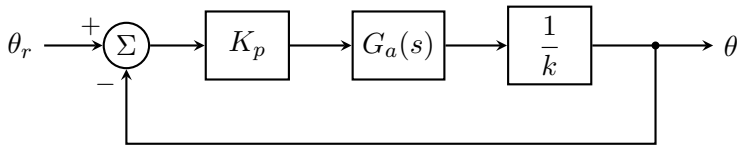


Figure 2: Fig. 4.26(a) with the inner summing junction and tachometer loop combined into the single block $G_a(s)$.

$$G_a(s) = \frac{KK_m}{s(1+\tau_m s)} \frac{1}{1 + k_t s \frac{KK_m}{s(1+\tau_m s)}} \tag{1}$$

$$= \frac{KK_m}{s(1+\tau_m s)} \frac{s(1+\tau_m s)}{s(1+\tau_m s) + k_t s KK_m} \tag{2}$$

$$= \frac{KK_m}{s(1+\tau_m s) + KK_m k_t s} \tag{3}$$

$$\frac{\theta_a}{\theta_r} = \frac{K_p \frac{1}{k} \frac{KK_m}{s(1+\tau_m s)+KK_m k_t s}}{1 + K_p \frac{1}{k} \frac{KK_m}{s(1+\tau_m s)+KK_m k_t s}} \tag{4}$$

$$= \frac{K_p KK_m}{k\left(s(1+\tau_m s) + KK_m k_t s\right) + K_p KK_m} \tag{5}$$

$$= \frac{K_p KK_m}{k\tau_m s^2 + k(1 + KK_m k_t)s + K_p KK_m} \tag{6}$$

$$\frac{\theta_b}{\theta_r} = \frac{\frac{K'}{s(1+\tau_m s)}}{1 + (1 + k_t' s)\frac{K'}{s(1+\tau_m s)}} \tag{7}$$

$$= \frac{K'}{s(1 + \tau_m s) + (1 + k_t' s)K'} \tag{8}$$

$$= \frac{K'}{\tau_m s^2 + (1 + K'k_t')s + K'} \tag{9}$$

$$\frac{K_p K K_m}{k\tau_m s^2 + k(1 + K K_m k_t)s + K_p K K_m} = \frac{K'}{\tau_m s^2 + (1 + K'k_t')s + K'} \tag{10}$$

$$\frac{K_p K K_m/k}{\tau_m s^2 + (1 + K K_m k_t)s + K_p K K_m/k} = \frac{K'}{\tau_m s^2 + (1 + K'k_t')s + K'} \tag{11}$$

$$K' = \frac{K_p K K_m}{k} \tag{12}$$

$$K'k_t' = \frac{K_p K K_m}{k} k_t' = K K_m k_t \tag{13}$$

$$k_t' = \frac{k k_t}{K_p} \tag{14}$$

Problem 2.B

$$E_{cl} = \theta_r - \theta_b \tag{15}$$

$$= \left(1 - \frac{\theta_b}{\theta_r}\right)\theta_r \tag{16}$$

$$= \left(1 - \frac{K'}{\tau_m s^2 + (1 + K'k_t')s + K'}\right)\theta_r \tag{17}$$

$$= \left(\frac{\tau_m s^2 + (1 + K'k_t')s + K' - K'}{\tau_m s^2 + (1 + K'k_t')s + K'}\right)\theta_r \tag{18}$$

$$= \left(\frac{s(\tau_m s + 1 + K'k_t')}{\tau_m s^2 + (1 + K'k_t')s + K'}\right)\theta_r \tag{19}$$

Check against a step function:

$$e_{ss} = \lim_{s\to 0} s\,\frac{1}{s}\,\frac{s(\tau_m s + 1 + K'k_t')}{\tau_m s^2 + (1 + K'k_t')s + K'} \tag{20}$$

$$= \lim_{s\to 0} \frac{s(\tau_m s + 1 + K'k_t')}{\tau_m s^2 + (1 + K'k_t')s + K'} \tag{21}$$

$$= \frac{0}{K'} = 0 \tag{22}$$

Check against a ramp function:

$$e_{ss} = \lim_{s\to 0} s\,\frac{1}{s^2}\,\frac{s(\tau_m s + 1 + K'k'_t)}{\tau_m s^2 + (1 + K'k'_t)s + K'} \tag{23}$$

$$= \lim_{s\to 0} \frac{\tau_m s + 1 + K'k'_t}{\tau_m s^2 + (1 + K'k'_t)s + K'} \tag{24}$$

$$= \frac{1 + K'k'_t}{K'} \tag{25}$$

The system is type 1.

$$K_v = \frac{1}{e_{ss}} = \frac{K'}{1 + K'k'_t} \tag{26}$$

Problem 2.C

From equation (12) we have

$$K' = \frac{K_p K K_m}{k} \tag{12}$$

and from equation (14) we have

$$k'_t = \frac{k k_t}{K_p} \tag{14}$$

$$K_v = \frac{K'}{1 + K'k'_t} \tag{27}$$

$$= \frac{\frac{K_p K K_m}{k}}{1 + \frac{K_p K K_m}{k}\frac{k k_t}{K_p}} \tag{28}$$

$$= \frac{\frac{K_p K K_m}{k}}{1 + K K_m k_t} \tag{29}$$

$$= \frac{K_p K K_m}{k(1 + K K_m k_t)} \tag{30}$$

K_v decreases as positive k_t grows in magnitude.

Problem 3 — Working with a control loop and poles and zeros

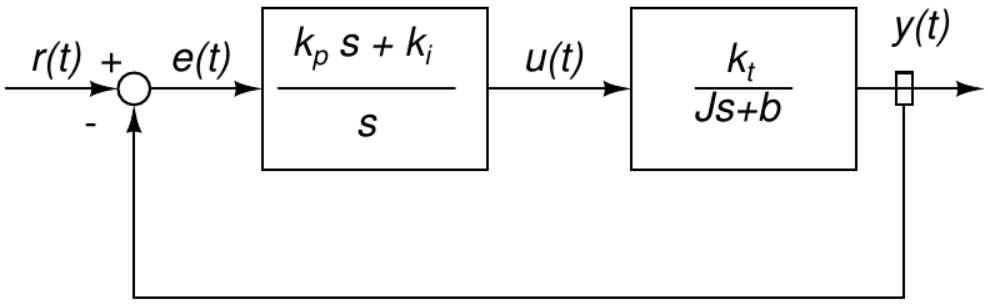


Figure 3: Block diagram of motor with control.

Problem 3.A

$$G_1 = \frac{k_p s + k_i}{s} \tag{31}$$

$$G_2 = \frac{k_t}{Js + b} \tag{32}$$

$$Y(s) = \frac{G_1 G_2}{1 + G_1 G_2} R(s) \tag{33}$$

$$= \frac{\frac{k_p s + k_i}{s} \frac{k_t}{Js+b}}{1 + \frac{k_p s + k_i}{s} \frac{k_t}{Js+b}} R(s) \tag{34}$$

$$= \frac{k_t(k_p s + k_i)}{s(Js + b) + k_t(k_p s + k_i)} R(s) \tag{35}$$

$$= \frac{k_t(k_p s + k_i)}{Js^2 + (b + k_t k_p)s + k_t k_i} R(s) \tag{36}$$

$$T_{ry}(s) = \frac{Y(s)}{R(s)} = \frac{k_t(k_p s + k_i)}{Js^2 + (b + k_t k_p)s + k_t k_i} \tag{37}$$

Problem 3.B

$$T_{ry}(s) = \frac{(1.8)\big((1.5)s + (0.4)\big)}{(15)s^2 + \big((2.0) + (1.8)(1.5)\big)s + (1.8)(0.4)} \tag{38}$$

$$= \frac{2.7s + 0.72}{15s^2 + (2.0 + 2.7)s + 0.72} \tag{39}$$

$$= \frac{2.7s + 0.72}{15s^2 + 4.7s + 0.72} \tag{40}$$

Problem 3.C

Code: hw_07_3C.m

```m
% Poles and zeros of T_ry(s) = (2.7s + 0.72) / (15s^2 + 4.7s + 0.72)
num = [2.7 0.72];
den = [15 4.7 0.72];

zeros_Try = roots(num)
poles_Try = roots(den)
```

Output: output/hw_07_3C.txt

```
zeros_Try = -0.2667
poles_Try =

  -0.1567 + 0.1532i
  -0.1567 - 0.1532i
```

Problem 3.D

$$p_{1,2} = -\alpha \pm j\omega \tag{41}$$

$$\omega_n = \sqrt{\alpha^2 + \omega^2} \tag{42}$$

$$\rho = \frac{\alpha}{\omega_n} \tag{43}$$

Code: `hw_07_3D.m`

```
den = [15 4.7 0.72];
p = roots(den);

alpha = -real(p(1))
omega = abs(imag(p(1)))

omega_n = sqrt(alpha^2 + omega^2)
rho = alpha/omega_n

t_r = 1.8/omega_n

M_p = 100*exp(-pi*alpha/omega)

t_s = 4.6/alpha
```

Output: `output/hw_07_3D.txt`

```
alpha = 0.1567
omega = 0.1532
omega_n = 0.2191
rho = 0.7151
t_r = 8.2158
M_p = 4.0208
t_s = 29.362
```

D.1

$$t_r \approx \frac{1.8}{\omega_n} \tag{44}$$

$$\approx 8.22 \text{ s} \tag{45}$$

D.2

$$\rho = \frac{\alpha}{\omega_n} \approx 0.715 \tag{46}$$

D.3

$$M_p = 100\% \times e^{-\pi \frac{\alpha}{\omega}} = 100\% \times e^{-\frac{\pi\rho}{\sqrt{1-\rho^2}}} \tag{47}$$

D.4

$$t_s \approx \frac{4.6}{\alpha} = \frac{4.6}{\rho\omega_n} \tag{48}$$

$$\approx 29.4 \text{ s} \tag{49}$$

Problem 3.E

Code: hw_07_3E.m

```
if exist('OCTAVE_VERSION', 'builtin')
    pkg load control;
end

num = [2.7 0.72];
den = [15 4.7 0.72];
sys = tf(num, den)

step(sys)
info = stepinfo(sys, 'SettlingTimeThreshold', 0.01)
pole(sys)
zero(sys)
```

Output: output/hw_07_3E.txt

```
Transfer function 'sys' from input 'u1' to output ...

          2.7 s + 0.72
 y1:  ---------------------
      15 s^2 + 4.7 s + 0.72

Continuous-time model.
info =

  scalar structure containing the fields:

    RaiseTime = [](0x0)
    RiseTime = 6.6790
    TransientTime = 25.849
    SettlingTime = 25.849
    SettlingMin = 0.9000
    SettlingMax = 1.0748
    Overshoot = 7.4782
    Undershoot = 0
    Peak = 1.0748
    PeakTime = 14
    StaticGain = 1
    InitialJump = 0

ans =

  -0.1567 + 0.1532i
  -0.1567 - 0.1532i

ans = -0.2667
```

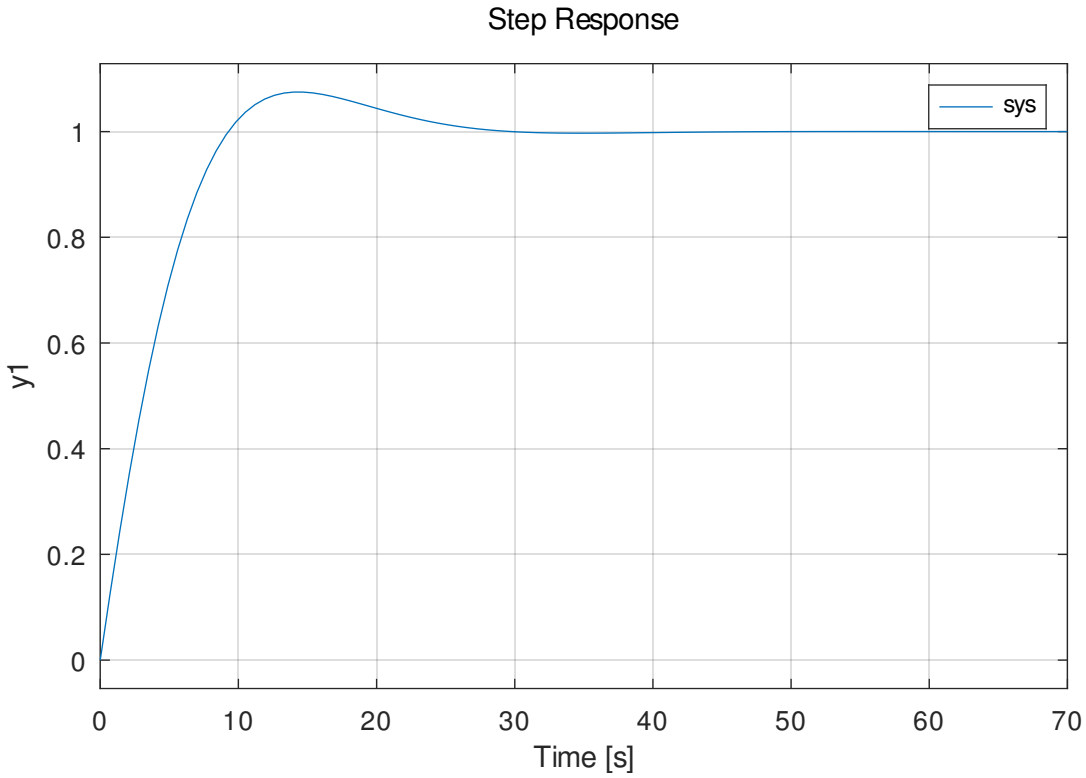


Figure 4: Step response of $T_{ry}(s)$.

Problem 3.F

(a) is the approximate method (i), which does not take the zeros into account.
(b) includes all of the system information in the plot, so those values are more precise (ii).

Problem 3.G

$$T_{ry}(s) = \frac{k_t(k_p s + k_i)}{Js^2 + (b + k_t k_p)s + k_t k_i} \tag{37}$$

$$T_{ry}(s) = \frac{(1.8)\big((5.0)s + (1.2)\big)}{(15)s^2 + \big((2.0) + (1.8)(5.0)\big)s + (1.8)(1.2)} \tag{50}$$

$$= \frac{9.0s + 2.16}{15s^2 + (2.0 + 9.0)s + 2.16} \tag{51}$$

$$= \frac{9.0s + 2.16}{15s^2 + 11s + 2.16} \tag{52}$$

Code: `hw_07_3G.m`

```
if exist('OCTAVE_VERSION', 'builtin')
    pkg load control;
end

num_G = [9 2.16];
den_G = [15 11 2.16];

sys_G = tf(num_G, den_G);

pole_G = pole(sys_G)
zero_G = zero(sys_G)

num_C = [2.7 0.72];
den_C = [15 4.7 0.72];

sys_C = tf(num_C, den_C);

pole_C = pole(sys_C)
zero_C = zero(sys_C)

pzmap(sys_G, sys_C)
legend('Part G poles', 'Part G zeros', 'Part C poles', 'Part C zeros', 'location', 'south')
grid on
```

Output: `output/hw_07_3G.txt`

```
pole_G =

  -0.3667 + 0.0978i
  -0.3667 - 0.0978i

zero_G = -0.2400
pole_C =

  -0.1567 + 0.1532i
  -0.1567 - 0.1532i

zero_C = -0.2667
```

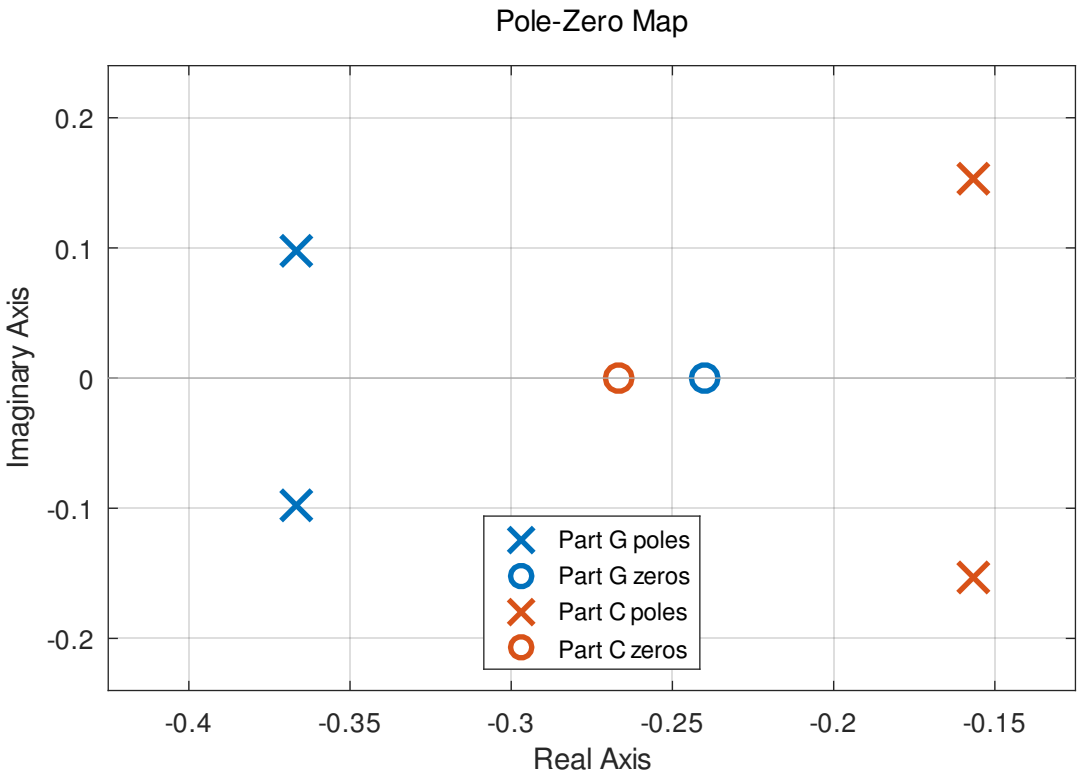


Figure 5: S-plane poles and zeros of $T_{ry}(s)$ for the original (part C) and new (part G) gains.

Problem 3.H

H.1

$$t_r \approx \frac{1.8}{\omega_n} \tag{44}$$

Given ω_n is the distance of the poles from the origin, I would expect t_r to be smaller (faster) with the new values, since the new poles are farther from the origin.

H.2

$$\rho = \frac{\alpha}{\omega_n} \tag{43}$$

I would expect the damping factor to be greater for the Part G equations, since the ratio of the real part to the total length of the vector from the origin to the poles is greater.

H.3

$$M_p = 100\% \times e^{-\pi \frac{\alpha}{\omega}} = 100\% \times e^{-\frac{\pi\rho}{\sqrt{1-\rho^2}}} \tag{47}$$

Since the ratio of the real part to the imaginary part is so much greater with the new values, I would expect the percent overshoot to be less with the new values.

H.4

$$t_s \approx \frac{4.6}{\alpha} = \frac{4.6}{\rho\omega_n} \tag{48}$$

I would expect the settling time to be lower, since the magnitude of the real part (α) is greater.

Problem 3.I

Code: `hw_07_3I.m`

```
if exist('OCTAVE_VERSION', 'builtin')
    pkg load control;
end

num_G = [9 2.16];
den_G = [15 11 2.16];

sys_G = tf(num_G, den_G);

num_C = [2.7 0.72];
den_C = [15 4.7 0.72];

sys_C = tf(num_C, den_C);

step_C = stepinfo(sys_C, 'SettlingTimeThreshold', 0.01)
damp_C = damp(sys_C)

step_G = stepinfo(sys_G, 'SettlingTimeThreshold', 0.01)
damp_G = damp(sys_G)
```

Output: `output/hw_07_3I.txt`

```
step_C =

  scalar structure containing the fields:

    RaiseTime = [](0x0)
    RiseTime = 6.6790
    TransientTime = 25.849
    SettlingTime = 25.849
    SettlingMin = 0.9000
    SettlingMax = 1.0748
    Overshoot = 7.4782
    Undershoot = 0
    Peak = 1.0748
    PeakTime = 14
    StaticGain = 1
    InitialJump = 0

damp_C =

   0.2191
   0.2191

step_G =

  scalar structure containing the fields:

    RaiseTime = [](0x0)
    RiseTime = 2.7344
    TransientTime = 14.761
    SettlingTime = 14.761
    SettlingMin = 0.9000
    SettlingMax = 1.0566
    Overshoot = 5.6587
    Undershoot = 0
    Peak = 1.0566
    PeakTime = 6.6000
    StaticGain = 1
    InitialJump = 0

damp_G =

   0.3795
   0.3795
```

The new values align with the expectations from part H.

Problem 4 — Text problem 4.7

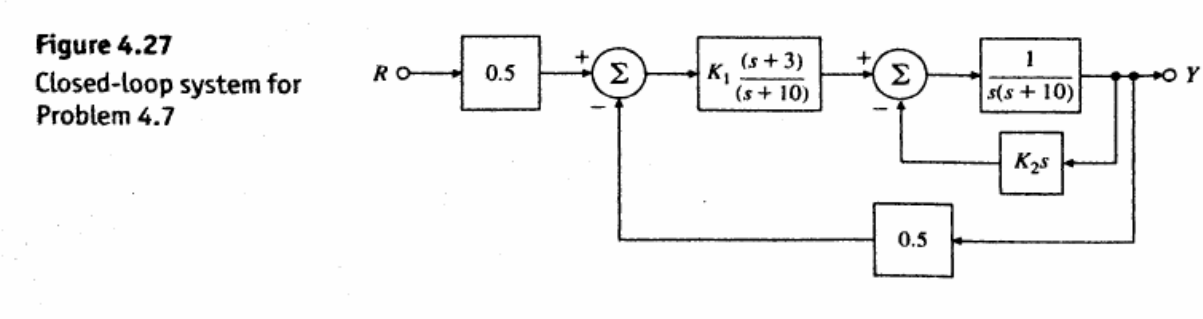


Figure 6: Text figure 4.27 (8th Ed.).

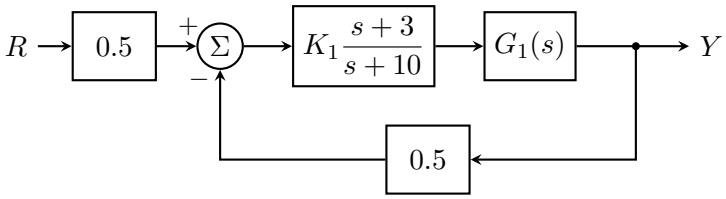


Figure 7: Fig. 4.27 with the inner summing junction and $K_2 s$ feedback loop combined into the single block $G_1(s)$.

$$G_1 = \frac{1}{s(s+10)} \frac{1}{1 + \frac{K_2 s}{s(s+10)}} \tag{53}$$

$$= \frac{1}{s(s+10) + K_2 s} \tag{54}$$

$$= \frac{1}{s(s+10+K_2)} \tag{55}$$

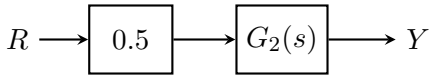


Figure 8: The remaining loop, $K_1 \frac{s+3}{s+10}$ and $G_1(s)$ with the 0.5 feedback, combined into the single block $G_2(s)$.

$$G_2 = \frac{K_1 \frac{s+3}{s+10} G_1}{1 + 0.5 K_1 \frac{s+3}{s+10} G_1} \tag{56}$$

$$= \frac{K_1 (s+3) G_1}{(s+10) + 0.5 K_1 (s+3) G_1} \tag{57}$$

$$= \frac{K_1 (s+3) \frac{1}{s(s+10+K_2)}}{(s+10) + 0.5 K_1 (s+3) \frac{1}{s(s+10+K_2)}} \tag{58}$$

$$= \frac{K_1 (s+3)}{s(s+10)(s+10+K_2) + 0.5 K_1 (s+3)} \tag{59}$$

$$\frac{Y}{R} = 0.5 G_2 \tag{60}$$

$$= \frac{0.5 K_1 (s+3)}{s(s+10)(s+10+K_2) + 0.5 K_1 (s+3)} \tag{61}$$

Problem 4.A

$$E = R - Y \tag{62}$$

$$= \left(1 - \frac{Y}{R}\right) R \tag{63}$$

$$= \left(1 - \frac{0.5 K_1 (s+3)}{s(s+10)(s+10+K_2) + 0.5 K_1 (s+3)}\right) R \tag{64}$$

$$= \left(\frac{s(s+10)(s+10+K_2) + 0.5 K_1 (s+3)}{s(s+10)(s+10+K_2) + 0.5 K_1 (s+3)} - \frac{0.5 K_1 (s+3)}{s(s+10)(s+10+K_2) + 0.5 K_1 (s+3)}\right) R \tag{65}$$

$$= \left(\frac{s(s+10)(s+10+K_2)}{s(s+10)(s+10+K_2) + 0.5 K_1 (s+3)}\right) R \tag{66}$$

$$e_{ss} = \lim_{s \to 0} s \frac{1}{s} \frac{s(s+10)(s+10+K_2)}{s(s+10)(s+10+K_2) + 0.5 K_1 (s+3)} \tag{67}$$

$$= \lim_{s \to 0} \frac{s(s+10)(s+10+K_2)}{s(s+10)(s+10+K_2) + 0.5 K_1 (s+3)} \tag{68}$$

$$= \frac{0}{1.5 K_1} = 0 \tag{69}$$

The steady-state tracking error is 0.

Problem 4.B

$$e_{ss} = \lim_{s\to 0} s\,\frac{1}{s^2}\,\frac{s(s+10)(s+10+K_2)}{s(s+10)(s+10+K_2)+0.5K_1(s+3)} \tag{70}$$

$$= \lim_{s\to 0} \frac{(s+10)(s+10+K_2)}{s(s+10)(s+10+K_2)+0.5K_1(s+3)} \tag{71}$$

$$= \frac{10(10+K_2)}{1.5K_1} \tag{72}$$

The system is type 1.

Problem 4.C

$$\frac{Y}{R} = \frac{0.5K_1(s+3)}{s(s+10)(s+10+K_2)+0.5K_1(s+3)} \tag{61}$$

$$= \frac{0.5K_1(s+3)}{s(s+10)(s+12)+0.5K_1(s+3)} \tag{73}$$

$$= \frac{0.5K_1(s+3)}{s^3+22s^2+120s+0.5K_1s+1.5K_1} \tag{74}$$

$$= \frac{0.5K_1s+1.5K_1}{s^3+22s^2+(120+0.5K_1)s+1.5K_1} \tag{75}$$

Code: `hw_07_4C.m`

```
if exist('OCTAVE_VERSION', 'builtin')
    pkg load control;
end

K_2 = 2;
Overshoot_target = 17;

% K_1 values are checked from low to high, so stop at the first one over target
for K_1_try = 1000:2000
    num = [.5*K_1_try 1.5*K_1_try];
    den = [1 22 120+0.5*K_1_try 1.5*K_1_try];

    sys_try = tf(num, den);
    sysinf_try = stepinfo(sys_try);

    if sysinf_try.Overshoot > Overshoot_target
        break;
    end

    K_1 = K_1_try;
    sysinf = sysinf_try;
end

K_1
sysinf

ess = 5 * 10*(10 + K_2) / (1.5 * K_1)
```

Output: `output/hw_07_4C.txt`

```
K_1 = 1177
sysinf =

  scalar structure containing the fields:

    RaiseTime = [](0x0)
    RiseTime = 0.063136
    TransientTime = 0.6256
    SettlingTime = 0.6256
    SettlingMin = 0.8769
    SettlingMax = 1.1699
    Overshoot = 16.991
    Undershoot = 0
    Peak = 1.1699
    PeakTime = 0.1346
    StaticGain = 1
    InitialJump = 0

ess = 0.3398
```

The largest integer gain that does not exceed 17% overshoot is $K_1 = 1177$.

$$e_{ss} = 5.0 \frac{10(10 + K_2)}{1.5K_1} \tag{76}$$

$$= 5.0 \frac{10(12)}{1.5(1177)} \tag{77}$$

$$= 0.340 \tag{78}$$